



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohd Razia Alangir Banu
Student Name	G. Rishitha
Roll Number	2520030101
Branch	CSE
Course Outcome	CO4 — Analyze and implement Dijkstra's, Bellman-Ford, and Floyd-Warshall short-path algorithms to compare their applicability and correctness across different graph types (weighted/unweighted, directed/undirected, negative-weight edges) in logistics and routing system case studies.

Case Study : SkillBridge – Learning Path Optimization using Shortest Path Algorithms

Work Done Summary:

In this case study, we implemented advanced shortest path algorithms — Dijkstra's Algorithm, Bellman-Ford Algorithm, Floyd-Warshall Algorithm, and Topological Sorting — for SkillBridge's online skill development and certification platform. The platform manages multiple courses, certifications, and prerequisite skill dependencies, requiring efficient path computation to guide students toward their certification goals.

The implementation modeled courses and skills as graph structures, where nodes represent courses and edges represent prerequisite relationships with weights based on difficulty, duration, or dependency importance. Dijkstra's algorithm was used to calculate the shortest learning path from a starting course to a target certification. Bellman-Ford was applied to handle weighted dependencies and detect any inconsistent negative dependency cycles. Floyd-Warshall was utilized to analyze all-skill relationships and compute shortest paths between all available courses. Topological sorting was implemented to determine the correct course completion order based on prerequisite constraints.

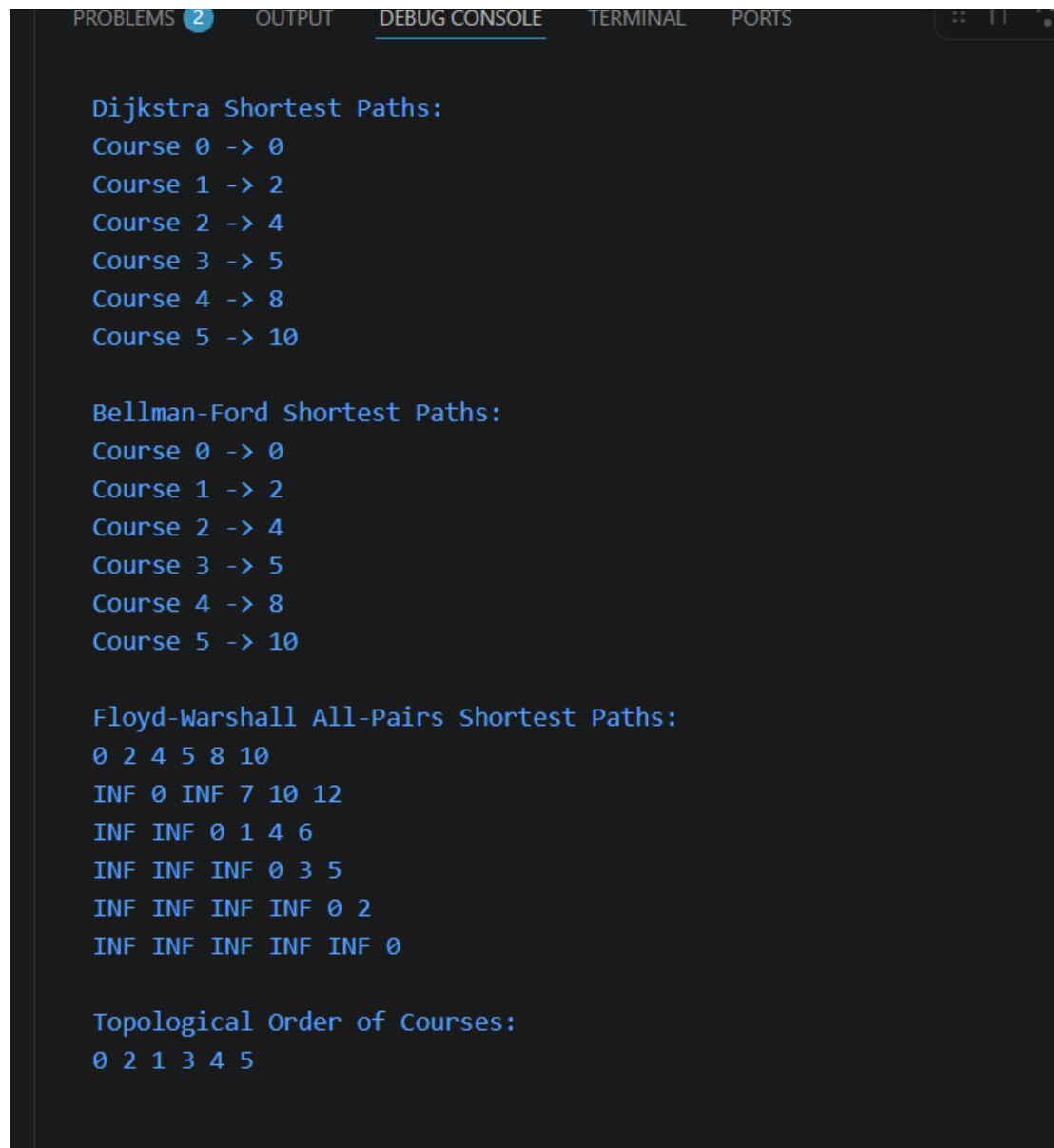
The system successfully optimized personalized learning paths, reduced unnecessary course selections, and improved career recommendation accuracy. This enhanced student learning efficiency by identifying the most effective and shortest paths toward certification completion.

Implementation Details:

1. Graph creation to represent courses, certifications, and prerequisite dependencies.
2. Weighted edge assignment based on course duration, complexity, and dependency.
3. Implementation of Dijkstra's algorithm for shortest certification path computation.
4. Implementation of Bellman-Ford algorithm for weighted skill dependency analysis.
5. Negative cycle detection for identifying invalid prerequisite loops.

6. Implementation of Floyd-Warshall algorithm for all-course relationship analysis.
 7. Topological sorting to generate valid course completion sequences.
 8. Performance comparison of shortest path algorithms on learning network graphs.
 9. Personalized recommendation generation based on shortest skill paths.
- Dijkstra's Algorithm: $O((V + E) \log V)$
 - Bellman-Ford Algorithm: $O(VE)$
 - Floyd-Warshall Algorithm: $O(V^3)$
 - Topological Sorting: $O(V + E)$

Result: Screenshots / Output:



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PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Dijkstra Shortest Paths:
Course 0 -> 0
Course 1 -> 2
Course 2 -> 4
Course 3 -> 5
Course 4 -> 8
Course 5 -> 10

Bellman-Ford Shortest Paths:
Course 0 -> 0
Course 1 -> 2
Course 2 -> 4
Course 3 -> 5
Course 4 -> 8
Course 5 -> 10

Floyd-Warshall All-Pairs Shortest Paths:
0 2 4 5 8 10
INF 0 INF 7 10 12
INF INF 0 1 4 6
INF INF INF 0 3 5
INF INF INF INF 0 2
INF INF INF INF INF 0

Topological Order of Courses:
0 2 1 3 4 5
```

Result and complexity:

1. Dijkstra's algorithm successfully generated the shortest learning path to certifications.
2. Bellman-Ford handled weighted skill dependencies and detected negative cycles correctly.
3. Floyd-Warshall provided complete relationship mapping between all courses and skills.
4. Topological sorting ensured proper prerequisite completion order.
5. The system optimized learning plans and improved student progression efficiency.
 - Dijkstra's Algorithm: $O((V + E) \log V)$
 - Bellman-Ford Algorithm: $O(VE)$
 - Floyd-Warshall Algorithm: $O(V^3)$

GitHub Link:

<https://github.com/rishi07-lab/DSA-2-CASE-STUDIES>

Note: The code and github link and screenshots have been uploaded in the above link . Please check it out there.

Student Signature	Faculty Signature
	